

BCA MATHEMATICS-I (BCA-0101)

Complete Unit-II Study Material

As per Himachal Pradesh University (HPU) BCA Semester-I Syllabus

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1 Rectangular Coordinates (Cartesian System)

1.1 Cartesian Coordinate System

The Cartesian coordinate system consists of two perpendicular lines:

- **X-axis:** Horizontal line
- **Y-axis:** Vertical line
- **Origin (O):** Point of intersection (0, 0)

1.2 Coordinates of a Point

Any point P in the plane is represented as (x, y) where:

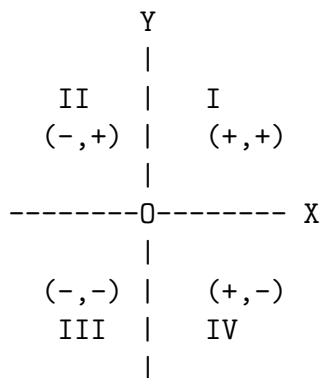
- **x-coordinate (Abscissa):** Distance from Y-axis
- **y-coordinate (Ordinate):** Distance from X-axis

Example: Point P(3, 4) means:

- 3 units from Y-axis (positive direction)
- 4 units from X-axis (positive direction)

1.3 Quadrants

Quadrant	x-coordinate	y-coordinate	Example
I	Positive (+)	Positive (+)	(3, 4)
II	Negative (-)	Positive (+)	(-3, 4)
III	Negative (-)	Negative (-)	(-3, -4)
IV	Positive (+)	Negative (-)	(3, -4)



1.4 Important Observations

- **Points on X-axis:** $(x, 0) \rightarrow$ y-coordinate is 0
- **Points on Y-axis:** $(0, y) \rightarrow$ x-coordinate is 0
- **Origin:** (0, 0)
- **Distance from X-axis:** $|y|$
- **Distance from Y-axis:** $|x|$

2 Distance Formula (Length of a Line Segment)

2.1 Distance Between Two Points

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2.2 Proof (Derivation)

Using Pythagoras theorem in right triangle:

$$PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2.3 Special Cases

Distance from Origin: For point $P(x, y)$, distance from origin $O(0, 0)$:

$$OP = \sqrt{x^2 + y^2}$$

Distance on X-axis: For points $(x_1, 0)$ and $(x_2, 0)$:

$$d = |x_2 - x_1|$$

Distance on Y-axis: For points $(0, y_1)$ and $(0, y_2)$:

$$d = |y_2 - y_1|$$

2.4 Applications

1. **Check if points are collinear:** If sum of distances between two pairs equals the third distance.
2. **Check if triangle is:** Equilateral, Isosceles, Scalene, or Right-angled.
3. **Check if quadrilateral is:** Square, Rectangle, Rhombus, or Parallelogram.

3 Section Formula (Section Ratio)

3.1 Internal Division

If point P(x, y) divides the line segment joining A(x₁, y₁) and B(x₂, y₂) in the ratio **m : n** internally, then:

$$x = \frac{mx_2 + nx_1}{m + n}, \quad y = \frac{my_2 + ny_1}{m + n}$$

$$\begin{array}{ccccc} \text{A} & \text{-----} & \text{P} & \text{-----} & \text{B} \\ (\text{x}_1, \text{y}_1) & & \text{m : n} & & (\text{x}_2, \text{y}_2) \end{array}$$

3.2 External Division

If point P(x, y) divides the line segment joining A(x₁, y₁) and B(x₂, y₂) in the ratio **m : n** externally, then:

$$x = \frac{mx_2 - nx_1}{m - n}, \quad y = \frac{my_2 - ny_1}{m - n}$$

$$\begin{array}{ccccc} \text{P} & \text{-----} & \text{A} & \text{-----} & \text{B} \\ (\text{x}, \text{y}) & & (\text{x}_1, \text{y}_1) & & (\text{x}_2, \text{y}_2) \\ & & \text{m : n (external)} & & \end{array}$$

3.3 Midpoint Formula

When P divides AB in the ratio **1 : 1** (internally):

$$x = \frac{x_1 + x_2}{2}, \quad y = \frac{y_1 + y_2}{2}$$

3.4 Centroid of a Triangle

The centroid (G) of triangle with vertices A(x₁, y₁), B(x₂, y₂), C(x₃, y₃):

$$G \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

3.5 Incenter of a Triangle

The incenter (I) of triangle with vertices A(x₁, y₁), B(x₂, y₂), C(x₃, y₃) and sides a, b, c:

$$I \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$$

where a = BC, b = CA, c = AB

4 Area of a Triangle

4.1 Formula

The area of triangle with vertices A(x_1, y_1), B(x_2, y_2), C(x_3, y_3):

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

4.2 Alternate Form

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

4.3 Condition for Collinearity

Three points A(x_1, y_1), B(x_2, y_2), C(x_3, y_3) are collinear if:

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

OR

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

4.4 Area of a Quadrilateral

For quadrilateral with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4):

$$\text{Area} = \frac{1}{2} |(x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1) - (y_1x_2 + y_2x_3 + y_3x_4 + y_4x_1)|$$

5 Equations of a Straight Line

5.1 Slope (Gradient)

The slope of a line passing through points (x_1, y_1) and (x_2, y_2) :

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \theta$$

where θ is the angle the line makes with positive X-axis.

Special Cases:

- Horizontal line: $m = 0$
- Vertical line: $m = \infty$ (undefined)
- Line at 45° : $m = 1$

5.2 Different Forms of Equation of a Line

5.2.1 1. Slope-Intercept Form

$$y = mx + c$$

where m = slope, c = y-intercept

5.2.2 2. Point-Slope Form

$$y - y_1 = m(x - x_1)$$

where (x_1, y_1) is a point on the line

5.2.3 3. Two-Point Form

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

5.2.4 4. Intercept Form

$$\frac{x}{a} + \frac{y}{b} = 1$$

where a = x-intercept, b = y-intercept

5.2.5 5. Normal Form

$$x \cos \alpha + y \sin \alpha = p$$

where p = perpendicular distance from origin, α = angle of normal

5.2.6 6. General Form

$$Ax + By + C = 0$$

Form	Equation	Conversion
Slope-Intercept	$y = mx + c$	$m = -A/B, c = -C/B$
General to Slope	$Ax + By + C = 0$	$y = (-A/B)x + (-C/B)$
Intercept from General	$x/(-C/A) + y/(-C/B) = 1$	$a = -C/A, b = -C/B$

5.3 Relationship Between Different Forms

5.4 Angle Between Two Lines

If slopes of two lines are m_1 and m_2 :

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

Conditions:

- **Parallel:** $m_1 = m_2$
- **Perpendicular:** $m_1 \times m_2 = -1$

5.5 Distance from a Point to a Line

Distance from point (x_1, y_1) to line $Ax + By + C = 0$:

$$d = \left| \frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}} \right|$$

5.6 Distance Between Two Parallel Lines

For lines $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$:

$$d = \left| \frac{C_2 - C_1}{\sqrt{A^2 + B^2}} \right|$$

5.7 Concurrency of Three Lines

Three lines:

- $L_1: A_1x + B_1y + C_1 = 0$
- $L_2: A_2x + B_2y + C_2 = 0$
- $L_3: A_3x + B_3y + C_3 = 0$

They are concurrent if:

$$\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = 0$$

6 Circles

6.1 Definition

A circle is the set of all points in a plane that are at a fixed distance (radius) from a fixed point (center).

6.2 Standard Equation

Circle with center (h, k) and radius r:

$$(x - h)^2 + (y - k)^2 = r^2$$

Circle with center at origin (0, 0):

$$x^2 + y^2 = r^2$$

6.3 General Equation

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Where:

- **Center:** (-g, -f)
- **Radius:** $r = \sqrt{g^2 + f^2 - c}$

6.4 Converting General to Standard Form

$$\begin{aligned} x^2 + y^2 + 2gx + 2fy + c &= 0 \\ (x + g)^2 + (y + f)^2 &= g^2 + f^2 - c \end{aligned}$$

6.5 Diameter Form

If A(x₁, y₁) and B(x₂, y₂) are endpoints of diameter:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

6.6 Condition for Circle

For equation $x^2 + y^2 + 2gx + 2fy + c = 0$:

Condition	Nature
$g^2 + f^2 - c > 0$	Real circle
$g^2 + f^2 - c = 0$	Point circle (radius = 0)
$g^2 + f^2 - c < 0$	Imaginary circle (no real locus)

6.7 Parametric Equation of Circle

For circle $x^2 + y^2 = r^2$:

$$x = r \cos \theta, \quad y = r \sin \theta$$

6.8 Tangent to a Circle

Tangent at point (x_1, y_1) on circle $x^2 + y^2 = r^2$:

$$xx_1 + yy_1 = r^2$$

Tangent at point (x_1, y_1) on circle $x^2 + y^2 + 2gx + 2fy + c = 0$:

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

6.9 Normal to a Circle

The normal at any point on a circle passes through the center.

6.10 Condition for Tangency

Line $y = mx + c$ is tangent to circle $x^2 + y^2 = r^2$ if:

$$c^2 = r^2(1 + m^2)$$

7 Practice Questions (Topic-wise)

7.1 Rectangular Coordinates

Q1. In which quadrant do the following points lie?

- (a) (3, -4)
- (b) (-2, 5)
- (c) (-1, -3)
- (d) (4, 2)

Answer: (a) IV (b) II (c) III (d) I

Q2. What are the coordinates of a point on X-axis at distance 5 units from origin?

Answer: (5, 0) or (-5, 0)

7.2 Distance Formula

Q3. Find the distance between points A(2, 3) and B(5, 7).

Answer:

$$\begin{aligned} AB &= \sqrt{(5-2)^2 + (7-3)^2} \\ &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} = 5 \text{ units} \end{aligned}$$

Q4. Find the distance of point P(3, 4) from origin.

Answer:

$$OP = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ units}$$

Q5. Show that the points A(2, 3), B(5, 7), and C(-1, -1) are collinear.

Answer:

$$\begin{aligned} AB &= \sqrt{(5-2)^2 + (7-3)^2} = \sqrt{25} = 5 \\ BC &= \sqrt{(-1-5)^2 + (-1-7)^2} = \sqrt{36 + 64} = 10 \\ AC &= \sqrt{(-1-2)^2 + (-1-3)^2} = \sqrt{9 + 16} = 5 \\ AB + AC &= 5 + 5 = 10 = BC \end{aligned}$$

$\therefore$ Points are collinear.

7.3 Section Formula

Q6. Find the coordinates of the point dividing the line segment joining A(2, 3) and B(5, 9) in the ratio 2:1 internally.

Answer:

$$\begin{aligned} x &= \frac{2 \times 5 + 1 \times 2}{2 + 1} = \frac{10 + 2}{3} = 4 \\ y &= \frac{2 \times 9 + 1 \times 3}{2 + 1} = \frac{18 + 3}{3} = 7 \end{aligned}$$

$\therefore P(4, 7)$

Q7. Find the centroid of triangle with vertices A(2, 3), B(4, 7), C(6, 5).

Answer:

$$\begin{aligned} G &= \left(\frac{2+4+6}{3}, \frac{3+7+5}{3} \right) \\ &= \left(\frac{12}{3}, \frac{15}{3} \right) = (4, 5) \end{aligned}$$

7.4 Area of Triangle

Q8. Find the area of triangle with vertices A(1, 2), B(3, 4), C(5, 6).

Answer:

$$\begin{aligned} \text{Area} &= \frac{1}{2} |1(4-6) + 3(6-2) + 5(2-4)| \\ &= \frac{1}{2} |1(-2) + 3(4) + 5(-2)| \\ &= \frac{1}{2} |-2 + 12 - 10| \\ &= \frac{1}{2} |0| = 0 \end{aligned}$$

$\therefore$ Points are collinear.

Q9. Find the area of triangle with vertices A(2, 3), B(5, 7), C(1, 4).

Answer:

$$\begin{aligned} \text{Area} &= \frac{1}{2} |2(7-4) + 5(4-3) + 1(3-7)| \\ &= \frac{1}{2} |2(3) + 5(1) + 1(-4)| \\ &= \frac{1}{2} |6 + 5 - 4| \\ &= \frac{1}{2} |7| = 3.5 \text{ square units} \end{aligned}$$

7.5 Straight Line

Q10. Find the slope of line passing through A(2, 3) and B(5, 9).

Answer:

$$m = \frac{9-3}{5-2} = \frac{6}{3} = 2$$

Q11. Find the equation of line with slope 2 and y-intercept 3.

Answer: $y = 2x + 3$

Q12. Find the equation of line passing through (2, 3) with slope 4.

Answer:

$$\begin{aligned} y - 3 &= 4(x - 2) \\ y - 3 &= 4x - 8 \\ y &= 4x - 5 \end{aligned}$$

Q13. Show that lines $2x + 3y = 5$ and $4x + 6y = 10$ are parallel.

Answer:

Slope of first: $2x + 3y = 5$

$$y = -\frac{2}{3}x + \frac{5}{3}, \quad m_1 = -\frac{2}{3}$$

Slope of second: $4x + 6y = 10$

$$y = -\frac{2}{3}x + \frac{5}{3}, \quad m_2 = -\frac{2}{3}$$

$m_1 = m_2$, so lines are parallel.

7.6 Circles

Q14. Find the equation of circle with center $(2, 3)$ and radius 5.

Answer: $(x-2)^2 + (y-3)^2 = 25$

Q15. Find the center and radius of circle $x^2 + y^2 - 4x - 6y - 12 = 0$.

Answer:

$$2g = -4 \rightarrow g = -2$$

$$2f = -6 \rightarrow f = -3$$

$$\text{Center} = (-g, -f) = (2, 3)$$

$$r = \sqrt{g^2 + f^2 - c} = \sqrt{4 + 9 + 12} = \sqrt{25} = 5$$

Q16. Find the tangent to circle $x^2 + y^2 = 25$ at point $(3, 4)$.

Answer: $3x + 4y = 25$

8 HPU Exam Important Questions (Unit-II Only)

Based on HPU BCA-0101 Previous Year Papers

8.1 Section-A (5 Marks Each)

Q1. Find the distance between points (2, 3) and (5, 7).

Q2. Find the coordinates of point dividing the line segment joining A(1, 2) and B(4, 8) in ratio 2:3 internally.

Q3. Find the area of triangle with vertices (2, 3), (4, 7), and (6, 9).

Q4. Find the equation of line passing through (2, 3) with slope 5.

Q5. Find the center and radius of circle $x^2 + y^2 - 6x - 8y + 9 = 0$.

8.2 Section-B (5 Marks Each)

Q6. For what value of k, the line passing through points (k, 4), (1, 2) is parallel to the line passing through points (3, -1), (2, -2)?

Q7. Find the equation of the line passing through the point (3, 4) and perpendicular to the line $2x + 3y = 5$.

Q8. Find the distance of point (1, 2) from the line $3x + 4y + 7 = 0$.

8.3 Section-C (10 Marks Each)

Q9. Prove that the points (2, 3), (4, 7), and (6, 11) are collinear. Also find the equation of the line passing through them.

Q10. Find the equation of the circle whose center is (3, -2) and which passes through the point (1, 4).

Q11. Find the equation of the circle passing through the points (1, 2), (3, 4), and (5, 8).

8.4 Section-D (10 Marks Each)

Q12. A line passes through the point (2, 3) and makes an angle of 45° with the positive X-axis. Find its equation and the intercepts it makes on the axes.

Q13. Find the tangent to the circle $x^2 + y^2 = 25$ at point (3, 4). Also find the normal at that point.

8.5 Section-E (Compulsory - 1 Mark Each)

Q14. Write the distance formula between two points (x_1, y_1) and (x_2, y_2) .

Q15. What is the midpoint formula?

Q16. Write the centroid formula of a triangle.

Q17. What is the slope of a horizontal line?

Q18. What is the condition for two lines to be perpendicular?

Q19. Write the standard equation of a circle with center (h, k) and radius r.

Q20. What is the radius of circle $x^2 + y^2 = 16$?

Q21. What is the equation of the X-axis?

Q22. What is the equation of the Y-axis?

Q23. Find the slope of line $2x + 3y = 6$.

Q24. What is the distance of point (3, 4) from origin?

9 Quick Revision Formulas

9.1 Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

9.2 Section Formula (Internal)

$$x = \frac{mx_2 + nx_1}{m + n}, \quad y = \frac{my_2 + ny_1}{m + n}$$

9.3 Section Formula (External)

$$x = \frac{mx_2 - nx_1}{m - n}, \quad y = \frac{my_2 - ny_1}{m - n}$$

9.4 Midpoint Formula

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

9.5 Centroid Formula

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

9.6 Area of Triangle

$$A = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

9.7 Slope

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \theta$$

9.8 Equation Forms

Form	Equation
Slope-Intercept	$y = mx + c$
Point-Slope	$y - y_1 = m(x - x_1)$
Two-Point	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
Intercept	$x/a + y/b = 1$
General	$Ax + By + C = 0$

9.9 Angle Between Lines

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

9.10 Parallel Condition

$$m_1 = m_2$$

9.11 Perpendicular Condition

$$m_1 \times m_2 = -1$$

9.12 Distance from Point to Line

$$d = \left| \frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}} \right|$$

9.13 Circle Equations

Form	Equation
Center-Radius	$(x-h)^2 + (y-k)^2 = r^2$
Center at Origin	$x^2 + y^2 = r^2$
General	$x^2 + y^2 + 2gx + 2fy + c = 0$
Center from General	$(-g, -f)$
Radius from General	$r = \sqrt{g^2 + f^2 - c}$
Diameter Form	$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$

9.14 Tangent to Circle

$$xx_1 + yy_1 = r^2$$

9.15 Condition for Tangency (Line $y = mx + c$ to $x^2 + y^2 = r^2$)

$$c^2 = r^2(1 + m^2)$$

Exam Tips for Unit-II

1. **Distance Formula** - Most important, used in many problems
2. **Section Formula** - Internal & External division - Very common in exams
3. **Area of Triangle** - Direct formula and collinearity condition
4. **Equation of Line** - All 6 forms, especially Slope-Intercept and General
5. **Circle** - Center and radius from general equation

All the Best for Your HPU BCA Semester-I Mathematics Examination!

— End of Unit-II Notes —